Distribution Theory

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Basically, a *distribution* can be seen as a generalization of a function. The corresponding subject in functional analysis, called the *theory of distributions*, was developed by L. SCHWARTZ in the midth of the 20th century to describe ideas and concepts of theoretical physics by an exact mathematical formalism. From the functional analysis point of view, distribution theory relies mainly on the concepts of locally convex topological linear space, continuous linear functional and adjoint linear operator.

1. Continuous Linear Operators

As for normed spaces, the notion of *continuity* is of fundamental interest for linear operators between topological linear spaces. Let V and W be topological linear spaces with underlying field $\mathbb{R}$ or $\mathbb{C}$. Then a linear operator $A: V \to W$ is continuous if and only if it is continuous in $0 \in V$ or in any other point $x \neq 0$. But continuity of A is not equivalent to boundedness in this more general setting.

The space V is called *continuously embedded* into the space W, if there is an injective and continuous linear operator $\iota: V \to W$. In addition, if this mapping is surjective and both spaces are Fréchet, then the *open mapping theorem* claims that ι^{-1} is continuous, too, hence both mappings ι and ι^{-1} are *linear homeomorphisms* between these spaces (notation: $V \cong W$).

The concept of *adjoint operator* is useful to generalize the domains of linear partial differential operators and Fourier transformation to the very extensive spaces of distributions. Let V and W be locally convex and V^*, W^* the dual spaces of them. Given a continuous linear operator $A: V \to W$, each linear functional $f \in W^*$ gives rise to a linear functional $g \in V^*$ by the composition $g := f \circ A$. The assignment $f \mapsto g$ induces a continuous mapping $A^* : W^* \to V^*$, called the *transposed operator* of A. One says that the continuous linear functional g is the *pullback* of f along the operator A.

The adjoint operator A^* is also linear. If V, W are locally convex linear spaces and $A: V \to W$ is continuous, then A^* is injective if

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and only if $\mathfrak{R}(A)$ is dense in W. If A is bijective, then A^* is bijective, too, and its inverse A^{*-1} is the adjoint of A^{-1}.

Finally, if V and W are Fréchet spaces, then the solvability of the linear equation $Ax = y$ is related to certain properties of the operator A^*: in fact, the operator $A : V \to W$ is surjective iff A^* is injective and $\mathfrak{R}(A^*)$ is closed in V^*.

2. Distributions

The space $C_c^\infty(G)$ of *test functions* is of fundamental meaning in distribution theory. Recall that this space forms a linear subspace of $C^\infty(G)$ consisting of smooth functions with *compact* support

$$\operatorname{supp}(\phi) := \overline{\{x \in G : \phi(x) \neq 0\}}.$$

There is neither a countable family of seminorms making $C_c^\infty(G)$ into a Fréchet space nor there is a metric making $C_c^\infty(G)$ into a F-space.

But it is still possible to define a locally convex topology $\mathcal{T}$ on $C_c^\infty(G)$ such that this space is complete in the sense of a topological linear space. Instead of saying how the open sets of this topology $\mathcal{T}$ look like, we define $\mathcal{T}$ by means of a suitable concept of convergence, known as $\mathcal{D}$-convergence in $C_c^\infty(G)$.

A sequence (ϕ_n) in $C_c^\infty(G)$ is said to be convergent to $\phi \in C_c^\infty(G)$, if there is a compact set $K \subset G$ such that

(D-1) $\operatorname{supp}(\phi_n) \subset K$ for all $n \in \mathbb{N}$,

(D-2) for each multi-index α the sequence $(\partial^\alpha \phi_k)$ converges to the function $\partial^\alpha \phi$ *uniformly* in K,

the linear space $C_c^\infty(G)$ equipped with the topology induced by $\mathcal{D}$-convergence is usually denoted $\mathcal{D}(G)$. We have $\mathcal{D}(G) \hookrightarrow \mathcal{E}(G)$, that is, the space $\mathcal{D}(G)$ is continuously embedded into the space $\mathcal{E}(G)$ with respect to their locally convex topologies.

The benefit of the function space $C_c^\infty(G)$ relies on the fact that, due to the smoothness and compact support property of its elements, the integral

$$F_f(\phi) := \int_G f\,\phi\,dx, \qquad \phi \in C_c^\infty(G) \tag{2.1}$$

is well defined for a multiplicity of real or complex-valued functions f. The largest class of functions for which this integral has a reasonable meaning, is the class $L^1_{loc}(G)$ of *locally integrable functions*, that is,

$f \in L^1_{loc}(G)$ is Lebesgue-integrable on each compact subset $K \subset G$. This space incorporates a lot of other well-known function spaces, for example all the spaces $L^p(G)$ with $p \geq 1$, the space $L^\infty(G)$ of essentially bounded functions and the space $C(G)$ of continuous functions. Moreover, if μ is a positive measure on G, the integral

$$F_\mu(\phi) := \int_G \phi \, d\mu, \quad \phi \in C_c^\infty(G) \tag{2.2}$$

especially makes sense for all the Radon measures μ on G, since such measures are finite on compact sets by definition. Both mappings $\phi \mapsto F_f$ and $\phi \mapsto F_\mu$ as defined in (2.1) and (2.2) act linearly on $C_c^\infty(G)$ and hence give rise to linear functionals. But both of them are also continuous on the topological linear space $\mathcal{D}(G)$.

Let us denote each continuous linear functional $F : \mathcal{D}(G) \to \mathbb{C}$ a *Schwartz distribution*. We call $\mathcal{D}'(G)$ the dual space of $\mathcal{D}(G)$, so $\mathcal{D}'(G)$ is the linear space of Schwartz distributions on the domain G. One may also define Schwartz distributions with respect to $\mathcal{D}$-convergence, when applied to test functions: a linear functional F on the space $C_c^\infty(G)$ is denoted a Schwartz distribution, if $F(\phi_n) \to 0$ for any $\mathcal{D}$-convergent sequence (ϕ_n) converges to the zero function.

As already mentioned, all linear functionals derived from locally integrable functions and Radon measures are continuous and therefore Schwartz distributions. Indeed, the space $\mathcal{D}'(G)$ is extremely extensive and contains nearly all functions being important in applications, amongst others the functions belonging to the well-known spaces $L^p(G)$, $L^\infty(G)$, $C(G)$, $C(\overline{G})$ and $C_0(G)$.

A distribution is said to be *regular*, if it is the continuous linear functional F_f induced by a locally integrable function f, otherwise it is called *singular*. Well-known examples of singular distributions are the *Dirac distributions*

$$\delta_a(\phi) := \phi(a) \quad \text{for all } \phi \in C_c^\infty(G) \qquad (a \in G).$$

Regarding the space $\mathcal{E}(G)$, which is the space $C^\infty(G)$ in the locally convex topology induced by the family of seminorms

$$|\phi|_{N,n} := \max_{x \in K_n} \{ |(\partial^\alpha \phi)(x)| : |\alpha| \leq N \} \qquad (N, n \in \mathbb{N})$$

for each $\phi \in C^\infty(G)$, where (K_n) is a sequence of compact subsets of G with

$$K_1 \subset K_2 \ldots \subset G, \qquad \bigcup_{n \in \mathbb{N}} K_n = G$$

each element $F \in \mathcal{E}'(G)$ is called a *distribution with compact support*. This notion is founded by the fact that for such F there is a compact set $K_F \subset G$, called the *support* of F such that $F\phi = 0$ for all smooth functions ϕ having support in $G \setminus K_F$.

The denotation for this class of distributions is motivated as follows: if $F_f \in \mathcal{E}'(G)$ is regular, then the function $f \in L^1_{loc}(G)$ related to F_f has compact support, too. On the other hand, each measurable function f with compact support of course gives rise to a compactly supported distribution $F_f \in \mathcal{E}'(G)$. Since it is always possible to increase a dual space by decreasing the underlying linear space, there is a continuous inclusion $\mathcal{E}'(G) \hookrightarrow \mathcal{D}'(G)$, which is dual to the mentioned embedding $\mathcal{D}(G) \hookrightarrow \mathcal{E}(G)$. So we basically may regard each compactly supported distribution as a Schwartz distribution.

3. Convolution and Tensor Products

The usage of distributions for the solution theory of linear partial differential equations can only be exhausted in full strength if the notion of *convolution* between elements of the spaces $\mathcal{D}'(\mathbb{R}^d)$ and $\mathcal{E}'(\mathbb{R}^d)$ has been introduced as far as possible.

The convolution $\phi * \psi$ of functions $\phi, \psi \in \mathscr{S}(\mathbb{R}^d)$ is defined by

$$(\phi * \psi)(x) := \int_{\mathbb{R}^d} \phi(y)\, \psi(x-y)\, dy$$

and always exists, hence the space of Schwartz functions forms an algebra with respect to convolution as multiplicative operation. Furthermore, convolution between a regular distribution $F_f \in \mathcal{D}'(\mathbb{R}^d)$ and a test function $\phi \in \mathcal{D}(\mathbb{R}^d)$ is similarly defined:

$$(F_f * \phi)(x) := \int_{\mathbb{R}^d} f(y)\, \phi(x-y)\, dy.$$

More general, the convolution $F * G$ of distributions $F, G \in \mathcal{D}'(\mathbb{R}^d)$ exists, if the *support condition* is fulfilled, meaning that for each compact subset $K \subset \mathbb{R}^d$ the set

$$K_{F,G} := \{(x,y) : x \in \operatorname{supp}(F), y \in \operatorname{supp}(G), x+y \in K\} \subset \mathbb{R}^d \times \mathbb{R}^d$$

is compact, too. In this case we may define the convolution $F * G$ according to

$$(F * G)\, \phi := \iint\limits_{K_{F,G}} F(x)\, G(y)\, \phi(x+y)\, dx\, dy, \quad \phi \in C_c^\infty(\mathbb{R}^d).$$

The support condition is sufficient but not necessary for the existence of the convolutional distribution $F*G$. Nevertheless, the condition is always fulfilled if at least one of the distributions F, G is compactly supported. Notice also that convolution on the space $\mathcal{D}'(\mathbb{R}^d)$ is commutative, if exists.

Especially the convolution between $F \in \mathcal{D}'(G)$ and the special distribution δ_0 always exists and satisfies the equation $F * \delta_0 = F$, thus δ_0 serves as identity element of this operation. But convolution is also associative, hence for distributions $F, G, H \in \mathcal{D}'(\mathbb{R}^d)$ the equality

$$(F * G) * H \;=\; F * (G * H)$$

holds provided that the involved convolutions involved exist.

Besides the notion of convolution one also requires that of *tensor product* between distributions. Tensor products are useful to transform an inital value of a distributional differential equation into an inhomogeneous part of this equation. Based on the fact that for $\phi \in \mathcal{D}(\mathbb{R}^{d_1})$ and $\psi \in \mathcal{D}(\mathbb{R}^{d_2})$ the tensor product $\phi \otimes \psi$ defined by

$$(\phi \otimes \psi)(x, y) \;=\; \phi(x) \cdot \psi(y)\,, \quad (x, y) \in \mathbb{R}^{d_1+d_2}$$

belongs to the space $\mathcal{D}(\mathbb{R}^{d_1+d_2})$, there is a uniquely defined distribution $F \otimes G \in \mathcal{D}'(\mathbb{R}^{d_1+d_2})$ for $F \in \mathcal{D}'(\mathbb{R}^{d_1})$ and $G \in \mathcal{D}'(\mathbb{R}^{d_2})$, which for all test functions $\phi \in \mathcal{D}(\mathbb{R}^{d_1})$ and $\psi \in \mathcal{D}(\mathbb{R}^{d_2})$ fulfills the property

$$F \otimes G\,(\phi \otimes \psi) \;=\; F(\phi) \cdot G(\psi)\,,$$

also denoted the *tensor product* of F and G. Performing the tensor product is a commutative and associative operation, that is

$$F \otimes G = G \otimes F, \qquad (F \otimes G) \otimes H \;=\; F \otimes (G \otimes H).$$

Let ∂^α be the elementary differential expression given by the multi-index α. Then ∂^α is a continuous linear operator $\partial^\alpha : \mathcal{D}(G) \to \mathcal{D}(G)$, due to the fact that the space $\mathcal{D}(G)$ is endowed with a very fine topology.

The action $\partial^\alpha_{\mathcal{D}}$ of the expression ∂^α on the distributional space $\mathcal{D}'(G)$ is defined by transposition of the *formally adjoint* expression $(\partial^\alpha)^* := (-1)^\alpha \partial^\alpha$, or equivalently

$$(\partial^\alpha_{\mathcal{D}} F)\,\phi \;:=\; F((\partial^\alpha)^* \phi) \;=\; (-1)^{|\alpha|}\, F(\partial^\alpha \phi) \tag{3.1}$$

for all $F \in \mathcal{D}'(G)$ and $\phi \in \mathcal{D}(G)$.

It should be noted that this definition is chosen in order to obtain compatibility with possible classical derivatives of *regular* distribu-

tions. In fact, if $F_f \in \mathcal{D}'(G)$ is the distribution generated by a differentiable function $f : G \to \mathbb{C}$ as defined in (2.1), then $\partial^\alpha_{\mathcal{D}} F_f = F_{\partial^\alpha f}$ holds.

The mapping $F \mapsto \partial^\alpha_{\mathcal{D}} F$ proves to be continuous on the Schwartz space $\mathcal{D}'(G)$. We also have the following compatibility rules between convolutions and tensor products:

$$\partial^\alpha_{\mathcal{D}}(F * G) \;=\; (\partial^\alpha_{\mathcal{D}} F) * G \;=\; F * (\partial^\alpha_{\mathcal{D}} G)\,,$$
$$\partial^\alpha_{\mathcal{D}}(F \otimes G) = (\partial^\alpha_{\mathcal{D}} F) \otimes G = F \otimes (\partial^\alpha_{\mathcal{D}} G).$$

We continue with the introduction of the *Schwartz space* $\mathscr{S}(\mathbb{R}^d)$, the linear space of all smooth functions $\phi \in C^\infty(\mathbb{R}^d)$ having the additional property

$$|\partial^\alpha \phi(x) \cdot x^\beta| \to 0 \;\text{ for } |x| \to \infty \text{ and all multi-indices } \alpha, \beta,$$

where $|x|$ denotes the Euclidean norm of $x \in \mathbb{R}^d$. A function $\phi \in \mathscr{S}(\mathbb{R}^d)$ is called *rapidly decreasing* or briefly a *Schwartz function*. Hence Schwartz functions and all their derivatives vanish at infinity faster then the reciprocal of any polynomial. The most famous example of a Schwartz function is the Hermite function

$$\mathcal{H}_0(x) \;:=\; e^{-\frac{|x|^2}{2}}, \quad x \in \mathbb{R}^d.$$

Rather similar to the space $C^\infty(G)$, one can find for the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ a family of seminorms as follows: define for $\phi \in C^\infty(\mathbb{R}^d)$ and multi-indices α, β

$$p_{\alpha,\beta}(\phi) \;:=\; \sup_{x \in \mathbb{R}^d} |\, x^\alpha \partial^\beta \phi(x) \,|. \tag{3.2}$$

Then ϕ is a Schwartz function iff $p_{\alpha,\beta}(\phi) < \infty$ holds for all multi-indices α, β. The collection of mappings $p_{\alpha,\beta} : \mathscr{S}(\mathbb{R}^d) \to \mathbb{R}$ defines a countable family of seminorms, which gives rise to a locally convex topology $\mathcal{T}_{\mathscr{S}}$ on $\mathscr{S}(\mathbb{R}^d)$. Moreover, since the Schwartz space is complete with respect to $\mathcal{T}_{\mathscr{S}}$, it actually forms a *Fréchet space*.

Comparing the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ with the function spaces $\mathcal{D}(\mathbb{R}^d)$ and $\mathcal{E}(\mathbb{R}^d)$, it is easy to see that there is the chain of inclusions

$$\mathcal{D}(\mathbb{R}^d) \;\hookrightarrow\; \mathscr{S}(\mathbb{R}^d) \;\hookrightarrow\; \mathcal{E}(\mathbb{R}^d).$$

Moreover, by comparing the locally convex topologies of these spaces, it turns out that these inclusions are actually continuous.

Similar to the spaces $\mathcal{D}(G)$, we introduce the concept of *tempered distribution* as a continuous linear functional $T : \mathscr{S}(\mathbb{R}^d) \to \mathbb{C}$. The space of tempered distributions is usually denoted $\mathscr{S}'(\mathbb{R}^d)$. A tempered distribution generalizes the class of bounded and slow-growing locally integrable functions.

The following criterion determines under which conditions a linear functional f on $\mathscr{S}(\mathbb{R}^d)$ forms a tempered distribution: this is the case if and only if there is a constant $C > 0$ and integers $m, l \in \mathbb{N}$ such that

$$|f(\phi)| \leq C \cdot \sum_{|\alpha| \leq l, |\beta| \leq m} p_{\alpha,\beta}(\phi) \quad \text{for all } \phi \in \mathscr{S}(\mathbb{R}^d),$$

where the functions $p_{\alpha,\beta}$ are the seminorms defined by (3.2).

Comparing the known kinds of distributional spaces, we obtain a chain of inclusions, which is dual to that of the base spaces:

$$\mathcal{E}'(\mathbb{R}^d) \subset \mathscr{S}'(\mathbb{R}^d) \subset \mathcal{D}'(\mathbb{R}^d).$$

Endowing each of these spaces with the so-called *weak*-topology*, these embeddings are continuous. Hence a distribution having compact support can also be regarded as a tempered distribution, which in turn gives rise to a Schwartz distribution.

4. Fourier Transforms in Distributional Spaces

In this section we shall elaborate that Fourier transformation can be extended to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions.

First notice that the Fourier transform $\hat{\phi}$ of a Schwartz function $\phi \in \mathscr{S}(\mathbb{R}^d)$ is well defined, since $\mathscr{S}(\mathbb{R}^d)$ is a linear subspace of the convolution algebra $L^1(\mathbb{R}^d, +, *)$. Moreover, Fourier transformation restricted to the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ forms a linear mapping $\mathscr{F} : \mathscr{S}(\mathbb{R}^d) \to \mathscr{S}(\mathbb{R}^d)$, which sends each function $\phi \in \mathscr{S}(\mathbb{R}^d)$ to a further function $\hat{\phi} \in \mathscr{S}(\mathbb{R}^d)$ according to the well-known formula

$$\hat{\phi}(x) = \mathscr{F}(\phi)(x) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i<\xi,x>} \phi(\xi)\, d\xi \quad (x \in \mathbb{R}^d). \quad (4.1)$$

The extraordinary meaning of Schwartz functions is founded by the fact that the Fourier transformation has excellent properties on the space $\mathscr{S}(\mathbb{R}^d)$. More precisely, besides the well-known features of Fourier transformation in $L^1(\mathbb{R}^d, +, *)$, the mapping $\mathscr{F}$ is *bijective* on $\mathscr{S}(\mathbb{R}^d)$. Hence the *inverse Fourier transform* exists for all $\phi \in \mathscr{S}(\mathbb{R}^d)$

and is described by the *Fourier inversion formula*

$$\check{\phi}(\xi) \;=\; \mathscr{F}^{-1}(\phi)(\xi) \;:=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\, \phi(x)\, dx \qquad (\xi \in \mathbb{R}^d).$$

We next extend Fourier transformation in $\mathscr{S}(\mathbb{R}^d)$ to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions by transposition, that is

$$(\mathscr{F}T)\,\phi \;:=\; T(\mathscr{F}\phi), \qquad \phi \in \mathscr{S}(\mathbb{R}^d),\; T \in \mathscr{S}'(\mathbb{R}^d).$$

This mapping is bijective in the space $\mathscr{S}'(\mathbb{R}^d)$ and even continuous with respect to the weak*-topology on $\mathscr{S}'(\mathbb{R}^d)$. The inverse Fourier transform of a tempered distribution T is represented by the formula

$$\mathscr{F}^{-1}(T)(\phi) \;:=\; T(\mathscr{F}^{-1}\phi) \qquad (\phi \in \mathbb{R}^d).$$

For example, the Fourier transform $\mathscr{F}(\delta_0)$ of the Dirac distribution $\delta_0 \in \mathcal{S}'(\mathbb{R})$ is the continuous linear functional

$$\mathscr{F}(\delta_0)(\phi) \;:=\; \delta_0(\mathscr{F}(\phi)) \;=\; \hat{\phi}(0) \;=\; \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \phi(x)\, dx$$

with $\phi \in \mathscr{S}(\mathbb{R})$, so $\mathscr{F}(\delta_0)$ is nothing but the distribution induced by the constant function $x \mapsto \frac{1}{\sqrt{2\pi}}$.

5. Fundamental Solutions

We now study the action of linear partial differential operators in distributional spaces. First recall that if $\psi \in C^\infty(G)$ and $F \in \mathcal{D}'(G)$, then ψF defined by

$$(\psi F)\phi \;:=\; F(\psi\phi), \qquad \phi \in C_c^\infty(G)$$

also belongs to $\mathcal{D}'(G)$. This implies that the linear differential operators $c_\alpha\, \partial^\alpha$ with $c_\alpha \in C^\infty(G)$ are acting on the space $\mathcal{D}'(G)$, since for $F \in \mathcal{D}'(G)$ the mapping $c_\alpha \partial^\alpha F$ is a distribution again. By taking linear combinations of such operators, we conclude that the action of a linear partial differential operator

$$\mathcal{A}(x,u) \;:=\; \sum_{|\alpha| \le 2m} c_\alpha(x)\; \partial^\alpha u(x)$$

having smooth coefficients and acting on the space $\mathcal{D}'(G)$ is well-defined, which of course is especially true in case of $G = \mathbb{R}^d$.

In the following we introduce the concept of *fundamental solution* for a linear partial differential operators $\mathcal{A}$ having smooth coeffi-

cients c_α. It is nothing but a distribution $\Phi_{\mathcal{A}} \in \mathcal{D}'(\mathbb{R}^d)$ that forms a solution of the differential equation $\mathcal{A}\Phi_{\mathcal{A}} = \delta_0$, where δ_0 is the Dirac distribution with respect to the origin $x = 0$.

It is useful to know a fundamental solution of $\mathcal{A}$ as the following computation shows: let $F \in \mathcal{D}'(\mathbb{R}^d)$ be any Schwartz distribution and consider the inhomogeneous differential equation $\mathcal{A}U = F$. If convolution between $\Phi_{\mathcal{A}}$ and F exists, then the equalities

$$\mathcal{A}(\Phi_{\mathcal{A}} * F) \;=\; (\mathcal{A}\Phi_{\mathcal{A}}) * F \;=\; \delta_0 * F \;=\; F$$

are valid, so the convolutional product $\Phi_{\mathcal{A}} * F$ solves the distributional differential equation $\mathcal{A}U = F$. Notice that there is a distributional solution U of this equation especially for all distributions F having compact support.

The fundamental solution of a linear differential operator need not be unique, which can be showed by the following example: let $U \in \mathcal{D}'(\mathbb{R}^d)$ be a solution of the homogeneous equation $\mathcal{A}U = 0$, then the distribution $\Phi_{\mathcal{A}} + U$ is a fundamental solution of $\mathcal{A}$, too.

Regarding the existence of fundamental solutions for linear differential operators, the MALGRANGE-EHRENPREIS *theorem* has been shown to be a milestone in PDE theory:

THEOREM 5.1. Each linear differential operator $\mathcal{A} \neq 0$ with constant coefficients has a fundamental solution in the distributional space $\mathcal{D}'(\mathbb{R}^d)$.

It should be mentioned that there are counterexamples which demonstrate that this theorem is not valid for differential operators whose coefficients are non-constant.

In the following we describe the fundamental solutions of three linear differential operators, namely the *Laplace operator*, the *heat operator* and the *wave operator*.

1. The fundamental solution Φ_Δ^d of the Laplacian Δ depends on the number of spatial coordinates. Since Δ is invariant with respect to translations and rotations, it is nearby to conjecture $\Phi_\Delta^d(x) = \phi_d(|x|)$. Indeed, the functions

$$\Phi_\Delta(x) \;=\; \begin{cases} \frac{1}{2}|x| & (d=1) \\ -\frac{1}{2\pi}\ln|x| & (d=2) \\ \frac{1}{(d-2)\,\omega_d}|x|^{2-d} & (d\geq 3) \end{cases}$$

are solutions of the differential equations $\Delta\Phi_\Delta^d = \delta_0$ for $d\in\mathbb{N}$, where ω_d is the Lebesgue-Borel measure of the d-dimensional unit sphere. Each other fundamental solution of the Laplacian is of the form $\Phi_\Delta^d + P_1$, where P_1 is any polynomial of degree less than two.

2. The fundamental solution of the *heat operator* $\mathcal{A} := \partial_t - \Delta u$ in $\mathbb{R}^d$ is given by the so-called *heat kernel* function

$$\Phi_d(x,t) \;=\; \begin{cases} \frac{1}{(4\pi t)^{d/2}}\cdot e^{-\frac{|x|^2}{4t}}, & x\in\mathbb{R}^d,\ t>0, \\ 0, & x\in\mathbb{R}^d,\ t\leq 0. \end{cases}$$

Each other fundamental solution of the heat operator adopts the form $\Phi_d + c$ ($c\in\mathbb{C}$), so the kernel of the heat operator purely consists of constant distributions.

3. When investigating the fundamental solutions of the hyperbolic *wave operator* $\Box := \partial_{tt} - \Delta u$, one has to distinguish between even and odd spatial dimensions.

Let H be the *Heaviside function*, defined by $H(t) := 0$ for $t<0$ and $H(t) := 1$ for $t\geq 0$. Then we obtain for $d\leq 3$

$$\Phi_\Box^d(x,t) \;=\; \begin{cases} \frac{1}{2}H(t-|x|) & (d=1) \\ \frac{1}{2\pi}\,(t^2-|x|^2)^{-1/2}\,H(t-|x|) & (d=2) \\ \frac{1}{2\pi}\,\delta(t^2-|x|^2)\,H(t) & (d=3). \end{cases}$$

It is a notable fact that all the higher fundamental solutions of the wave operator can be derived from the functions $\Phi_\Box^1$ and $\Phi_\Box^2$. Using the variable $r := |x|$, the functions $\Phi_\Box^{d+2}$ ($d\in\mathbb{N}$) are recursively given by

$$\Phi_\Box^{d+2}(t,r) \;=\; \frac{-1}{2\pi r}\cdot\frac{\partial}{\partial_r}\Phi_\Box^d(t,r), \quad d\geq 1.$$

To determine the support of a fundamental solution $\Phi_\Box^d$, we first define for fixed $x_0\in\mathbb{R}^d$ the *light cone*

$$C^+(x_0) \;:=\; \{(t,x) : t>0, |x-x_0|<t\}$$

and obtain the support of the fundamental solution according to

$$\operatorname{supp}(\Phi_{\Box}^{d}) = \begin{cases} \partial C^{+}(0), & d \text{ odd or } d \geq 3, \\ \overline{C^{+}(0)}, & d \text{ even or } d = 1. \end{cases}$$

A linear differential operator $\mathcal{A}$ is called *hypoelliptic*, if with respect to the linear differential equation $\mathcal{A}u = f$ the condition $f \in C^{\infty}(G)$ implies $u \in C^{\infty}(G)$. Restricting to differential operators with constant coefficients, hypoellipticity can be characterized by means of a feature of the fundamental solution $\Phi_{\mathcal{A}_c}$: $\mathcal{A}_c$ is hypoelliptic if and only if $\Phi_{\mathcal{A}_c} = F_f$, where F_f is a regular distribution with generating function f being indefinitely differentiable in every $x \neq 0$.

We next introduce the concept of *singular support* $\Sigma(F)$ for $F \in \mathcal{D}'(\mathbb{R}^d)$: it is the smallest subset $S \in \mathbb{R}^d$ such that F remains regular on $\mathbb{R}^d \setminus S$. Then hypoellipticity of linear differential operators with constant coefficients are characterized as follows:

THEOREM 5.2. A linear partial differential operator $\mathcal{A}_c$ with constant coefficients is hypoelliptic if and only if $\Sigma(\mathcal{A}_c F) = \Sigma(F)$ for all $F \in \mathcal{D}'(\mathbb{R}^d)$.

Knowing the fundamental solutions of the elliptic operator Δ and the parabolic heat operator $\partial_t - \Delta$, it turns out that both operators are hypoelliptic. But theorem 5.2 implies that the hyperbolic wave operator $\Box = \partial_{tt} - \Delta$ is not hypoelliptic.

6. Free Space Problems

To show the benefit of knowing the fundamental solution of a linear partial differential operator with constant coefficients, we consider free space problems ($G = \mathbb{R}^d$) related to the Laplacian, the heat and the wave operator.

1. The *Poisson equation* in the entire space $\mathbb{R}^d$ is given by

$$\Delta u = -4\pi f, \quad f \in C(\mathbb{R}^d). \tag{6.1}$$

A function $u : \mathbb{R}^d \to \mathbb{R}$ is called a *classical solution* of (6.1), if it is twice continuously differentiable and fulfills the differential equation in each $x \in \mathbb{R}^d$. However, $U \in \mathcal{D}'(\mathbb{R}^d)$ is said to be a *distributional solution* of the Poisson equation, if

$$\Delta U = -4\pi F, \quad F \in \mathcal{D}'(\mathbb{R}^d) \tag{6.2}$$

holds in the sense of distributions.

Let Φ_Δ be the fundamental solution of the Laplacian. If the convolutional product $\Phi_\Delta * F$ exists, which is especially true for each regular distribution F_f with generating function $f \in C_c(\mathbb{R}^d)$, then the solution of (6.2) is obtained by convolution, that is, $U = \Phi_\Delta * F$.

For the physically interesting case $d = 3$ and for $f \in C_c(\mathbb{R}^3)$ we get a classical solution u_f, called the *Newton potential* for the function f:

$$u_f(x) = \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|}\, dy, \quad x \in \mathbb{R}^3.$$

2. By the *heat equation* equation in $\mathbb{R}^d$ is meant the inital value problem

$$\begin{cases} u_t(t,x) - \Delta u(t,x) = f(t,x) & (x \in \mathbb{R}^d,\ t > 0) \\ u(0,x) = 0, & (x \in \mathbb{R}^d). \end{cases} \tag{6.3}$$

We first define the concept of distributional solution for problem (6.3): let $U_0 \in \mathcal{D}'(\mathbb{R}^d)$ and $F \in \mathcal{D}'(\mathbb{R}^{d+1})$ are Schwartz distributions satisfying $\operatorname{supp}(F) \subset [0,\infty) \times \mathbb{R}^d$, then $U \in \mathcal{D}'(\mathbb{R}^{d+1})$ is called a *distributional solution*, if $\operatorname{supp}(U) \subset [0,\infty) \times \mathbb{R}^d$ and

$$U_t - \Delta U = F + U_0 \otimes \delta_0, \tag{6.4}$$

where $\delta_0 \in \mathcal{D}'(\mathbb{R})$ is the Dirac distribution. Here the initial condition of the problem has been transformed into an inhomogeneous part of the differential equation.

The existence of a solution for (6.4) is ensured, if $U_0 \in \mathcal{E}'(\mathbb{R}^d)$ and $F \in \mathcal{E}'(\mathbb{R}^{d+1})$. In this case the solution $U \in \mathcal{D}'(\mathbb{R}^d)$ is given by convolution of the fundamental equation Φ_d with the inhomogeneous parts of (6.4),

$$U = \Phi_d * (F + U_0 \otimes \delta_0) = \Phi_d * F + \Phi_d * U_0.$$

Given a classical solution u of (6.3), the related Schwartz distribution F_u is a solution in the distributional sense. Conversely, let F_u be regular and solve (6.4), then u forms a classical solution, if some regularity conditions are valid:

a. U, U_0 and F are regular distributions with associated functions $u, u_0, f \in L^1_{loc}(G)$,
b. u_0 is continuous on $\mathbb{R}^d$ and f is continuous on $[0,\infty) \times \mathbb{R}^d$,

c. u is twice continuously differentiable on $[0,\infty)\times\mathbb{R}^d$.

3. The initial-value problem for the *wave equation* in $\mathbb{R}^d$ is given by

$$\begin{cases} u_{tt}(t,x)-\Delta u(t,x) = f(t,x) & (x\in\mathbb{R}^d,\, t>0) \\ u(0,x)=u_0(x) & (x\in\mathbb{R}^d) \\ u_t(0,x)=u_1(x) & (x\in\mathbb{R}^d. \end{cases} \tag{6.5}$$

As for the heat equation, we define for (6.5) the concept of distributional solution, where again both initial value conditions are transformed into inhomogeneous parts of the equation by means of tensor products.

Let $U_0\in\mathcal{D}'(\mathbb{R}^d)$, $U_1\in\mathcal{D}'(\mathbb{R}^d)$ and $F\in\mathcal{D}'(\mathbb{R}^{d+1})$ be distributions, then U is called a distributional solution of (6.5), if

$$U_{tt}-\Delta U = F+U_0\otimes\delta_0'+U_1\otimes\delta_0\,, \tag{6.6}$$

where δ_0 is the Dirac distribution for $t=0$ and δ_0' is the distributional derivative of δ_0 with respect to the variable t.

Since the support of the function $U_0\otimes\delta_0'+U_1\otimes\delta_0$ belongs to the set $\mathbb{R}^{d+1}$, there is to each distribution $F\in\mathcal{D}'(\mathbb{R}^{d+1})$ with support

$$\operatorname{supp}(F) = \{\,(x,t)\in\mathbb{R}^d\times\mathbb{R} : t\geq 0\,\}$$

a further distribution $U_F := \Phi_{\square}^d * (F+U_0\otimes\delta_0'+U_1\otimes\delta_0)\in\mathcal{D}'(\mathbb{R}^{d+1})$, which in fact solves the distributional differential equation (6.6).

Each classical solution of (6.5) is also a distributional solution. However, the reverse statement only holds under the conditions

a. U, U_0, U_1 and F are regular and generated by ordinary functions $u, u_0, u_1, f\in L^1_{loc}(G)$,
b. $u_0\in C^1(\mathbb{R}^d)$ and $u_1\in C(\mathbb{R}^d)$,
c. $f\in C([0,\infty)\times\mathbb{R}^d)$ and
d. $u\in C^2((0,\infty)\times\mathbb{R}^d)\cap C([0,\infty)\times\mathbb{R}^d)$.

7. Applications of the Fourier Transform

The usability of Fourier transformation mainly relies on the fact that elementary differential operators ∂_k, applied to Schwartz functions ϕ, appear as multiplication operators to the Fourier transform $\mathscr{F}(\phi)$ and via verce, that is, we always have

$$\partial_k\,\phi(x) \overset{\mathscr{F}}{\longmapsto} i\,\xi_k\cdot\hat{\phi}(\xi)\,, \qquad i\,x_k\phi(x) \overset{\mathscr{F}}{\longmapsto} \partial_k\,\hat{\phi}(\xi)\,,$$

for $k=1,\dots,d$, as one can verify by partial integration. For any linear differential operator $\mathcal{A}_c$ with constant coefficients c_α and symbol $S_{\mathcal{A}_c}$ we accordingly obtain

$$(\mathcal{A}_c\,\phi)(x) \overset{\mathscr{F}}{\longmapsto} S_{\mathcal{A}_c}(i\xi)\cdot\hat{\phi}(\xi).$$

This is nothing but the fact that all the differential operators $\mathcal{A}_c$ are Fourier multipliers. They operate on functions $\phi\in\mathscr{S}(\mathbb{R}^d)$ like multiplications with the symbols $S_{\mathcal{A}_c}$ on the Fourier transform $\mathscr{F}\phi$. Hence a linear differential operator $\mathcal{A}_c$ with constant coefficients is conjugated to its symbol function $S_{\mathcal{A}_c}$ in the sense of

$$\mathcal{A}_c\phi \;=\; (\mathscr{F}^{-1}\circ S_{\mathcal{A}_c}\circ\mathscr{F})\,\phi \;=\; \mathscr{F}^{-1}(S_{\mathcal{A}_c}\hat{\phi}) \quad \text{ for all } \phi\in\mathscr{S}(\mathbb{R}^d).$$

We want incident how Fourier transformation can be used to solve linear differential equations in the whole space $\mathbb{R}^d$ and governed by the Laplacian. The basic idea is to transform the differential equation either into a purely algebraic equation or into an ordinary differential equation. Once these are solved, inverse Fourier transformation is applied to get the solution of the original problem.

Two examples for these methods will be presented. In the first one, the bijectivity of Fourier transformation on the Schwartz space is used to solve a specific differential equation, while in the second one the simplified computation of convolution products after performing Fourier transformation is helpful to find fundamental solutions of the underlying differential operators.

1. The equation $\mathcal{A}_c\,u=f$ with $f\in\mathscr{S}(\mathbb{R}^d)$ and governed by a differential operator $\mathcal{A}_c$ having constant coefficents can be converted by applying Fourier transformation into the algebraic equation

$$S_{\mathcal{A}_c}(i\xi)\,\hat{u}(\xi) \;=\; \hat{f}(\xi) \qquad (\xi\in\mathbb{R}^d). \tag{7.1}$$

Now, if the symbol function $S_{\mathcal{A}_c}$ of $\mathcal{A}_c$ owns the property

$$S_{\mathcal{A}_c}(i\xi)\neq 0 \quad \text{ for all } \xi=(\xi_1,\dots,\xi_d)\in\mathbb{R}^d, \tag{7.2}$$

equation (7.1) can be resolved for $\hat{u}(\xi)$, and the solution u of $\mathcal{A}_c\,u=f$ is obtained by applying inverse Fourier transformation:

$$\hat{u}(\xi)=\hat{f}(\xi)\,/\,S_{\mathcal{A}_c}(i\xi)$$

and

$$u(x) \;=\; \frac{1}{(2\pi)^{d/2}}\int_{\mathbb{R}^d} e^{i\xi x}\cdot S^{-1}_{\mathcal{A}_c}(i\xi)\cdot\hat{f}(\xi)\,d\xi.$$

2. If (7.2) holds, the Fourier transform method can also be applied

to solve the distributional differential equation $\mathcal{A}_c U = F$ with $F \in \mathscr{S}'(\mathbb{R}^d)$. Since $\mathcal{A}_c$ is bijective on $\mathscr{S}(\mathbb{R}^d)$ and

$$P_{\mathcal{A}_c}(i\xi) \;=\; P_{\mathcal{A}_c^*}(-\xi) \quad \text{for all } \xi \in \mathbb{R}^d$$

holds, the formally adjoint operator $\mathcal{A}_c^*$ is bijective on $\mathscr{S}(\mathbb{R}^d)$ and so its adjoint operator $\mathcal{A}_c = (\mathcal{A}_c^*)^*$ is bijective on the dual space $\mathscr{S}'(\mathbb{R}^d)$. Thus, if (7.2) is valid, the equation $\mathcal{A}_c U = F$ is uniquely solvable for each tempered distribution F.

Especially for the differential operator $\Delta_\lambda := -\Delta + \lambda$ and $\lambda > 0$ we obtain

$$S_{\Delta_\lambda}(i\xi) = |\xi|^2 + \lambda \neq 0$$

for $\xi \in \mathbb{R}^d$, hence the solution u of $\Delta_\lambda u = f$ with $f \in \mathscr{S}(\mathbb{R}^d)$ is

$$u(x) \;=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\xi x} \cdot \frac{\hat{f}(\xi)}{|\xi|^2 + \lambda}\, d\xi, \quad (x \in \mathbb{R}^d).$$

Partial Fourier transformation is a method where the operator $\mathscr{F}$ is applied only to the spatial variables of the partial differential equation. The result is an ordinary differential equation with respect to the time variable t. Then solving this equation and applying inverse Fourier transformation leads to the solution of the original problem.

In the following we sketch how partial Fourier transformation is useful to solve the initial-value problem of the heat and wave equation in the entire space $\mathbb{R}^d$.

1. The initial-value problem of the heat equation is given by

$$\begin{cases} u_t - \Delta u = 0 & (t > 0) \\ u(0, x) \;=\; f(x) & (x \in \mathbb{R}). \end{cases}$$

Applying Fourier transformation to both sides of the differential equation with respect to the spatial variables $x_1, \ldots, x_d$, we obtain

$$\frac{\partial}{\partial t}\hat{u}(t,\xi) \;+\; \xi^2\, \hat{u}(t,\xi) \;=\; 0, \qquad \hat{u}(0,\xi) = \hat{f}(\xi), \tag{7.3}$$

where $\xi = (\xi_1, \ldots, \xi_d)$ are the spatial variables.

Then multiplying both sides of the first equation in (7.3) by $e^{|\xi|^2 t}$ leads to

$$\frac{\partial}{\partial t}\, e^{|\xi|^2 t}\, \hat{u}(t,\xi) \;=\; 0$$

with solution $\hat{u}(t,\xi) = \phi(\xi)\, e^{-|\xi|^2 t}$. The mapping $\xi \mapsto \phi(\xi)$ is the integration constant for each fixed point ξ. The second equation in (7.3) implies $\phi(\xi) = \hat{f}(\xi)$, hence the Fourier transformed solution $\hat{u}$ of the initial-value problem is nothing but

$$\hat{u}(t,\xi) = \hat{f}(\xi) \cdot e^{-|\xi|^2 t}. \tag{7.4}$$

The solution of the original equation is obtained by applying inverse Fourier transformation and the rule $\mathscr{F}^{-1}(f \cdot g) = \mathscr{F}^{-1}(f) * \mathscr{F}^{-1}(g)$ to each side of (7.4):

$$\begin{aligned} u(t,x) &= \mathscr{F}^{-1}(\hat{f}(\xi) \cdot e^{-|\xi|^2 t}) = \mathscr{F}^{-1}(\hat{f}(\xi)) * \mathscr{F}^{-1}(e^{-|\xi|^2 t}) \\ &= f(x) * \mathscr{F}^{-1}(e^{-|\xi|^2 t}) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-|x-y|^2/(4t)}\, f(y)\, dy. \end{aligned}$$

The function

$$H_t(x,y) := \frac{1}{\sqrt{4\pi t}}\, e^{-|x-y|^2/(4t)} = \Phi_{\partial_t - \Delta}(t, x-y)$$

is called a *heat kernel*, since it constitutes the kernel of the integral operator $\mathcal{H}_t : f \mapsto H_t * f$.

2. The method also enables to solve the initial-value problem for the wave equation

$$\begin{cases} u_{tt} - \Delta u = 0 & (t > 0), \\ u(0,x) = f(x) & (x \in \mathbb{R}^d), \\ u_t(0,x) = g(x) & (x \in \mathbb{R}^d). \end{cases}$$

Applying partial Fourier transformation, this equation becomes into

$$\hat{u}_{tt}(t,\xi) = |\xi|^2 \hat{u}(t,\xi), \qquad \hat{u}(0,\xi) = \hat{f}(\xi), \qquad \hat{u_t}(0,\xi) = \hat{g}(\xi),$$

and the general solution of the ordinary differential equation in phase space is

$$\hat{u}(t,\xi) = \hat{f}(\xi)\, \cos(t|\xi|) + \hat{g}(\xi) : \frac{\sin(t|\xi|)}{|\xi|}.$$

Unfortunately, performing inverse Fourier transformation to $\hat{u}$ for any $d > 1$ is not an easy task. But for $d = 1$ we obtain

$$u_1(x,t) = \mathscr{F}^{-1}\left(\hat{f}(\xi) \cdot \cos(|\xi|)\right) = \frac{1}{2}\,[f(x+t) + f(x-t)],$$

$$u_2(x,t) \;=\; \mathscr{F}^{-1}\left(\hat{g}(\xi)\cdot\frac{\sin(|\xi|)}{|\xi|}\right) \;=\; \frac{1}{2}\int_{-t}^{+t} g(x+s)\,ds,$$

thus the solution of the wave equation is given by $u = u_1 + u_2$, known as the D'ALEMBERT *solution*

$$u(t,x) \;=\; \frac{1}{2}\left[f(x+t) \;+\; f(x-t)\right] \;+\; \frac{1}{2t}\int_{-t}^{+t} g(x+s)\,ds.$$

References

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